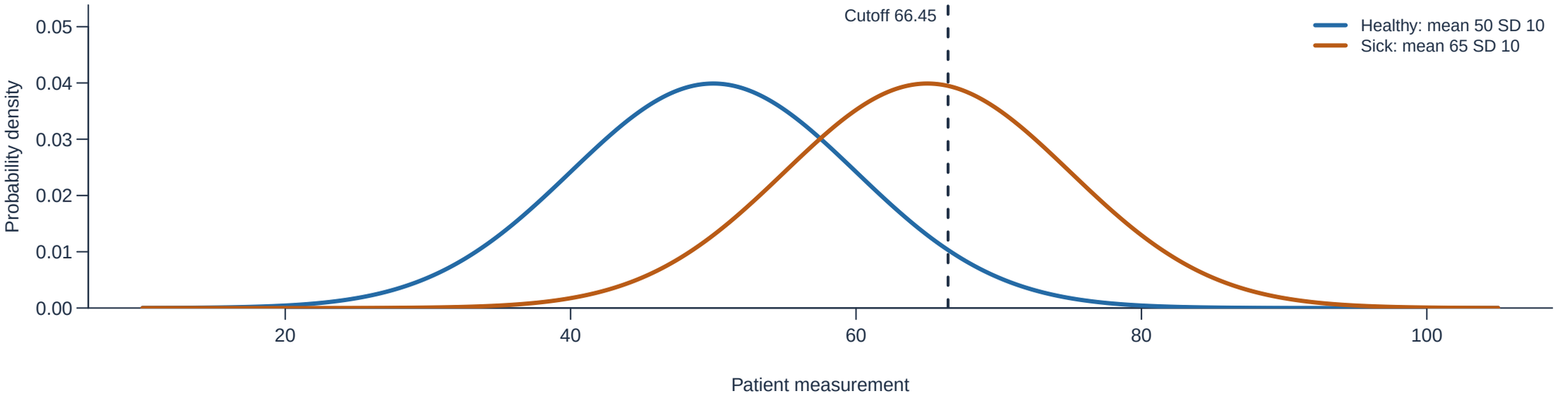
One measurement, two possible classification errors

Null hypothesis: healthy. Higher values trigger a sick classification.

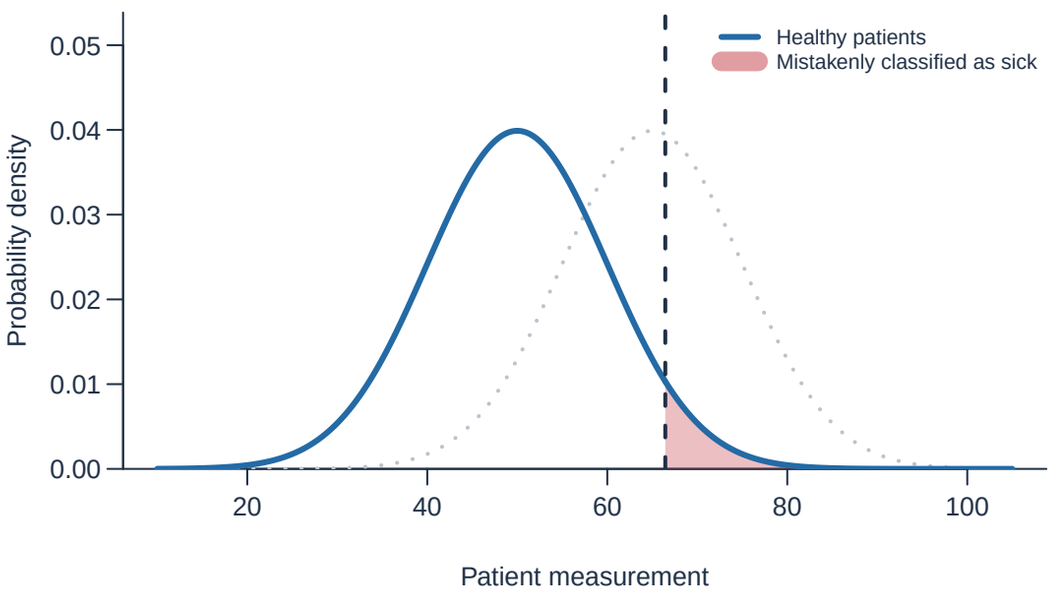
Two overlapping populations

Below cutoff: classify as healthy | At or above cutoff: classify as sick



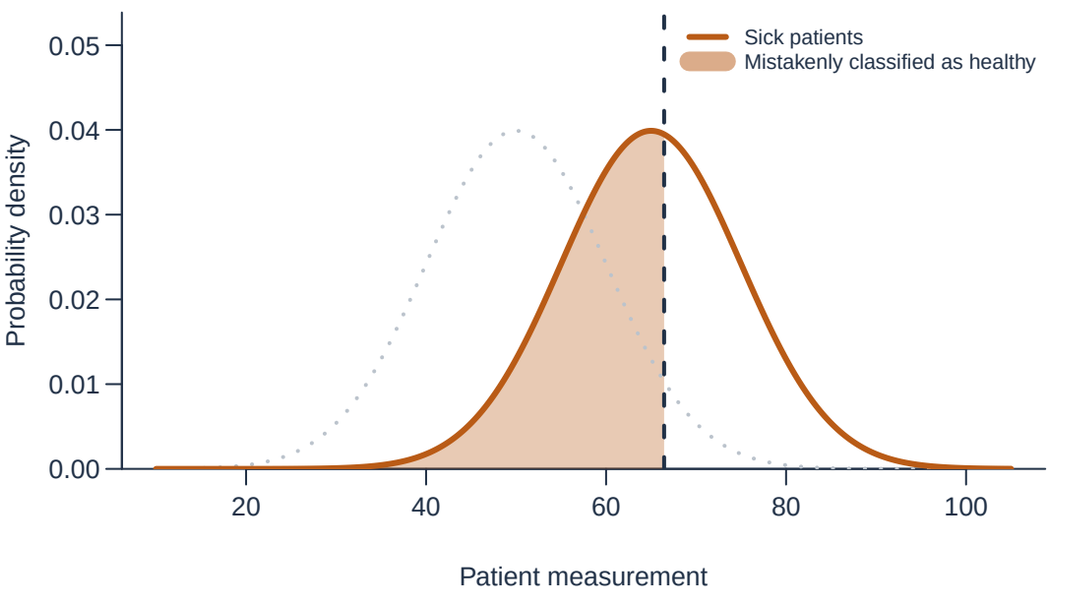
Alpha: false positives 5.0%

Area under the HEALTHY curve to the right of the cutoff



Beta: false negatives 55.8%

Area under the SICK curve to the left of the cutoff



Each shaded area is a probability within its own patient group.

These error rates do not give the probability that a particular patient is sick.